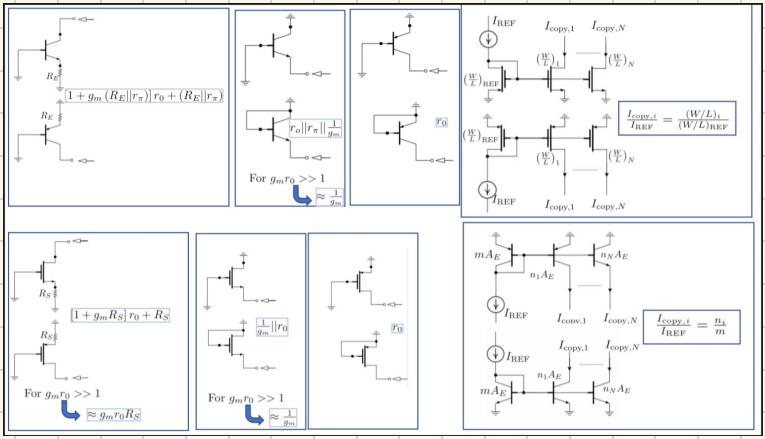
ELG 3136

Lecture 1

Finding Rout

$A_V = -G_m R_{out}$

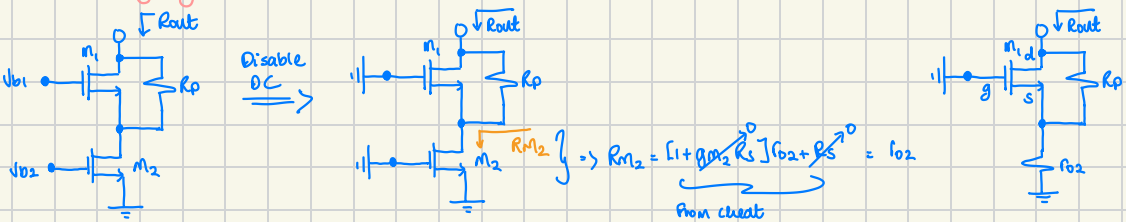
$G_m = \left.\frac{i_{out}}{v_{in}}\right|_{v_{out}=0}$



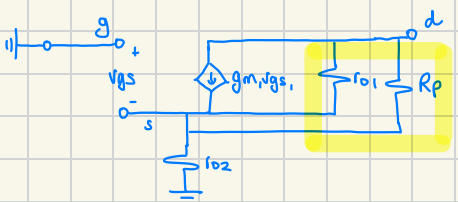
Rules when finding Rout

- Disable all sources
- Replace all transistors with resistors
- Combine → BOOM → Rout.

Example: finding only Rout of circuit below

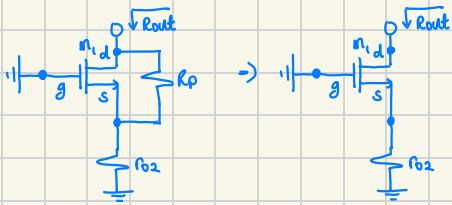


Draw small-signal model for intuition but don't do analysis



→ parallel: let $r_{O1}' = r_{O1} // R_D$

Now go back to previous schematic and ignore R_D: since we have $r_{O1}' = r_{O1} // R_D$



RMB we have r_{O1}' and not r_{O1}

Using cheat sheet: $R_{out} = [1+g_{m_1}R_S]r_{O1}' + R_S$, $R_S = r_{O2}$

$\Rightarrow R_{out} = [1+g_{m_1}r_{O2}]r_{O1}' + r_{O2} = [1+g_{m_1}r_{O2}](r_{O1} // R_D) + r_{O2}$

$\Rightarrow \boxed{R_{out} = [1+g_{m_1}r_{O2}](r_{O1} // R_D) + r_{O2}}$

Approx: $g_m r_O >> 1$

$R_{out} = [1+g_{m_1}r_{O2}](r_{O1} // R_D) + r_{O2} = g_{m_1}r_{O2}(r_{O1} // R_D) + r_{O2} = [g_{m_1}\underbrace{(r_{O1} // R_D)}_{=r_{O1}'} + 1]r_{O2} = g_{m_1}(r_{O1} // R_D)r_{O2}$

$\Rightarrow \boxed{R_{out} \approx g_{m_1}(r_{O1} // R_D)r_{O2}}$